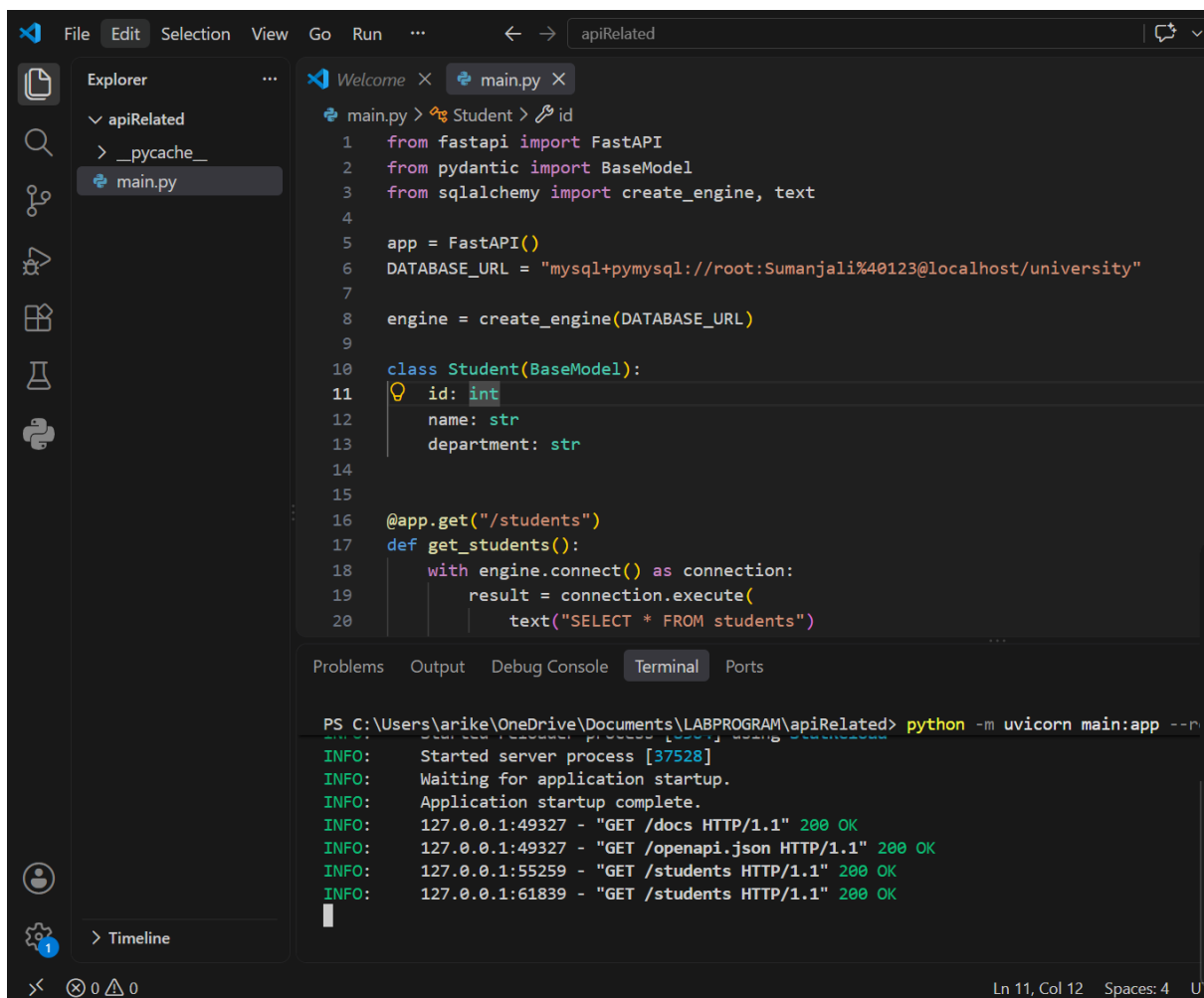


SKILL WEEK-5

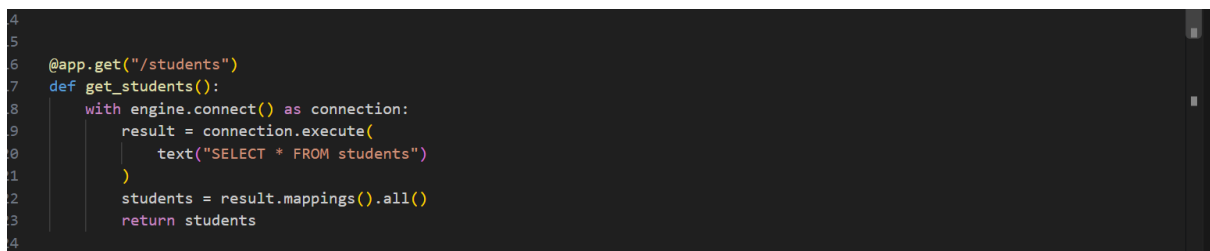


The screenshot shows the Visual Studio Code editor interface. The Explorer panel on the left shows a project named 'apiRelated' with a file 'main.py'. The main editor displays the code for 'main.py', which imports FastAPI, Pydantic, and SQLAlchemy. It sets up a database engine and a FastAPI application. A Pydantic model 'Student' is defined with fields 'id', 'name', and 'department'. A GET endpoint '/students' is implemented, which queries the database for all students. The bottom panel shows the Terminal output, indicating the server has started successfully and is receiving requests.

```
main.py > Student > id
1 from fastapi import FastAPI
2 from pydantic import BaseModel
3 from sqlalchemy import create_engine, text
4
5 app = FastAPI()
6 DATABASE_URL = "mysql+pymysql://root:Sumanjali%40123@localhost/university"
7
8 engine = create_engine(DATABASE_URL)
9
10 class Student(BaseModel):
11     id: int
12     name: str
13     department: str
14
15
16 @app.get("/students")
17 def get_students():
18     with engine.connect() as connection:
19         result = connection.execute(
20             text("SELECT * FROM students")
21         )
22     students = result.mappings().all()
23     return students
```

PS C:\Users\arique\OneDrive\Documents\LABPROGRAM\apiRelated> python -m uvicorn main:app --port 8000

INFO: Started server process [37528]
INFO: Waiting for application startup.
INFO: Application startup complete.
INFO: 127.0.0.1:49327 - "GET /docs HTTP/1.1" 200 OK
INFO: 127.0.0.1:49327 - "GET /openapi.json HTTP/1.1" 200 OK
INFO: 127.0.0.1:55259 - "GET /students HTTP/1.1" 200 OK
INFO: 127.0.0.1:61839 - "GET /students HTTP/1.1" 200 OK



This is a close-up view of the code for the '/students' endpoint. It shows the database connection, the execution of a SQL query, and the retrieval of student data from the database results.

```
@app.get("/students")
def get_students():
    with engine.connect() as connection:
        result = connection.execute(
            text("SELECT * FROM students")
        )
    students = result.mappings().all()
    return students
```

default

GET **/students** Get Students

Parameters

No parameters

Execute Clear

Responses

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8000/students' \
  -H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:8000/students
```

Server response

Code	Details
200	Response body [

FastAPI - Swagger UI

127.0.0.1:8000/docs#/default/get_students_students_get

Server response

Code	Details
200	Response body [{ "id": 101, "name": "Ravi", "department": "CSE" }]

Response headers

```
content-length: 45
content-type: application/json
date: Wed, 09 Sep 2026 10:16:58 GMT
server: uvicorn
```

Responses

Code	Description	Links
200	Successful Response	No links

Media type

application/json

Controls Accept header:

Example Value | Schema

"string"

Limit to 1000 rows

```
1 • CREATE DATABASE university;
2 • USE university;
3 • CREATE TABLE students (
4     id INT PRIMARY KEY,
5     name VARCHAR(100),
6     department VARCHAR(100)
7 );
8 • SELECT * FROM students;
```

MySQL Workbench

Local instance MySQL80 x MySQL Model x Local instance MySQL80 (uni... x

File Edit View Query Database Server Tools Scripting Help

Navigator week_2_practical SQL File 2 x SQLAdditions

MANAGEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System Va
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs

Administration

Information

No object selected

Result Grid Filter Rows: Edit: Export/Import:

	id	name	department
▶	101	Ravi	CSE
•	NULL	NULL	NULL

students 3 x Apply Revert Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
6	15:46:04	CREATE TABLE students (id INT PRIMARY KEY, ...	Error Code: 1050. Table 'students' already exists	0.016 sec
7	15:46:11	INSERT INTO students (id, name, department) VALUE...	1 row(s) affected	0.015 sec
8	15:46:14	SELECT * FROM students LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

```
24
25
26 @app.post("/students")
27 def create_student(student: Student):
28     with engine.begin() as connection:
29         connection.execute(
30             text("""
31                 INSERT INTO students (id, name, department)
32                 VALUES (:id, :name, :department)
33             """),
34             {
35                 "id": student.id,
36                 "name": student.name,
37                 "department": student.department
38             }
39         )
40
41     return {
42         "message": "Student created",
43         "student": student
44     }
```

FastAPI - Swagger UI

127.0.0.1:8000/docs#/default/create_student_students_post

POST /students Create Student

Parameters

No parameters

Request body **required**

application/json

Edit Value | Schema

```
{
  "id": 102,
  "name": "Priya",
  "department": "ECE"
}
```

Execute Clear

FastAPI - Swagger UI

127.0.0.1:8000/docs#/default/create_student_students_post

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8000/students' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
    "id": 102,
    "name": "Priya",
    "department": "ECE"
  }'
```

Request URL

http://127.0.0.1:8000/students

Server response

Code	Details
200	Response body <pre>{ "message": "Student created", "student": { "id": 102, "name": "Priya", "department": "ECE" } }</pre> Response headers <pre>content-length: 84 content-type: application/json date: Thu, 10 Sep 2026 03:59:17 GMT server: uvicorn</pre>

Responses

Code	Description
------	-------------

Links

MySQL Workbench

Local instance MySQL80 x MySQL Model x Local instance MySQL80 (uni... x

File Edit View Query Database Server Tools Scripting Help

Navigator: week_2_practical Skill_week-5 x SQLAdditions

MANAGEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System Variables
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs

Administration

Information: No object selected

Result Grid

id	name	department
101	Ravi	CSE
102	Priya	ECE
NULL	NULL	NULL

students 4 x Apply Revert Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
7	15:46:11	INSERT INTO students (id, name, department) VALUE...	1 row(s) affected	0.015 sec
8	15:46:14	SELECT * FROM students LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
9	09:31:13	SELECT * FROM students LIMIT 0, 1000	2 row(s) returned	0.032 sec / 0.000 sec

Object Info Session

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

```

44 }
45 @app.put("/students/{student_id}")
46 def update_student(student_id: int, student: Student):
47     with engine.begin() as connection:
48         connection.execute(
49             text("""
50                 UPDATE students
51                 SET name = :name,
52                     department = :department
53                 WHERE id = :student_id
54             """),
55             {
56                 "student_id": student_id,
57                 "name": student.name,
58                 "department": student.department
59             }
60         )
61
62     return {
63         "message": "Student updated",
64         "student_id": student_id,
65         "student": student
66     }

```

FastAPI - Swagger UI

Ask Gemini

127.0.0.1:8000/docs#/default/update_student_students__student_id_put

default

GET /students Get Students

POST /students Create Student

PUT /students/{student_id} Update Student

Parameters

Cancel

Reset

Name

Description

student_id required

integer

(path)

101

Request body required

application/json

Edit Value

Schema

```
{
  "id": 101,
  "name": "Ravi Kumar",
  "department": "AIML"
}
```

FastAPI - Swagger UI

Ask Gemini

127.0.0.1:8000/docs#/default/update_student_students__student_id_put

```
curl -X 'PUT' \
  'http://127.0.0.1:8000/students/101' \
  -H 'accept: application/json' \
  -H 'content-type: application/json' \
  -d '{
    "id": 101,
    "name": "Ravi Kumar",
    "department": "AIML"
  }'
```

Request URL

http://127.0.0.1:8000/students/101

Server response

Code

Details

200

Response body

```
{
  "message": "Student updated",
  "student_id": 101,
  "student": {
    "id": 101,
    "name": "Ravi Kumar",
    "department": "AIML"
  }
}
```

Download

Response headers

```
content-length: 107
content-type: application/json
date: Thu, 10 Sep 2026 04:05:07 GMT
server: uvicorn
```

Responses

Code

Description

Links

200

Successful Response

No links

MySQL Workbench

Local instance MySQL80 x MySQL Model x Local instance MySQL80 (uni... x

File Edit View Query Database Server Tools Scripting Help

Navigator week_2_practical Skill_week-5

MANAGEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System Va
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs

Administration

Information

No object selected

Result Grid

id	name	department
101	Ravi Kumar	AIML
102	Priya	ECE

students 5 x

Apply Revert Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
8	15:46:14	SELECT * FROM students LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
9	09:31:13	SELECT * FROM students LIMIT 0, 1000	2 row(s) returned	0.032 sec / 0.000 sec
10	09:35:55	SELECT * FROM students LIMIT 0, 1000	2 row(s) returned	0.015 sec / 0.000 sec

Object Info Session

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

```

64         "student_id": student_id,
65         "student": student
66     }
67     @app.delete("/students/{student_id}")
68     def delete_student(student_id: int):
69         with engine.begin() as connection:
70             connection.execute(
71                 text("""
72                     DELETE FROM students
73                     WHERE id = :student_id
74                     """),
75                 {
76                     "student_id": student_id
77                 }
78             )
79
80         return {
81             "message": "Student deleted",
82             "student_id": student_id
83         }

```

Ln 83, Col 6 Spaces: 4 UTF-8 CRLF {} Python Python 3.14.7 Go Live

FastAPI - Swagger UI | Ask Gemini | 127.0.0.1:8000/docs#/default/delete_student_students_student_id_delete

DELETE /students/{student_id} Delete Student

Parameters

Cancel

Name	Description
student_id * required	
integer (path)	101

Execute Clear

Responses

Curl

```
curl -X 'DELETE' \
  'http://127.0.0.1:8000/students/101' \
  -H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:8000/students/101
```

FastAPI - Swagger UI | Ask Gemini | 127.0.0.1:8000/docs#/default/delete_student_students_student_id_delete

Code Details

200

Response body

```
{
  "message": "Student deleted",
  "student_id": 101
}
```

Download

Response headers

```
content-length: 46
content-type: application/json
date: Thu, 10 Sep 2026 04:09:47 GMT
server: uvicorn
```

Responses

Code	Description	Links
200	Successful Response	No links
	Media type: application/json	
	Controls Accept header:	
	Example Value: "string"	
422	Validation Error	No links
	Media type: application/json	
	Example Value: Schema	

MySQL Workbench

Local instance MySQL80 x MySQL Model x Local instance MySQL80 (uni... x

File Edit View Query Database Server Tools Scripting Help

Navigator week_2_practical Skill_week-5

MANAGEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System Va
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs

Administration

Information

No object selected

Result Grid

id	name	department
102	Priya	ECE
NULL	NULL	NULL

students 6 x Apply Revert Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
9	09:31:13	SELECT * FROM students LIMIT 0, 1000	2 row(s) returned	0.032 sec / 0.000 sec
10	09:35:55	SELECT * FROM students LIMIT 0, 1000	2 row(s) returned	0.015 sec / 0.000 sec
11	09:40:11	SELECT * FROM students LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

```
6 department VARCHAR(100)
7 };
8 SELECT * FROM students;
9 INSERT INTO students (id, name, department)
10 VALUES (101, 'Ravi', 'CSE');
11 SELECT * FROM students;
12 SELECT * FROM students;
13 SELECT * FROM students;
```

FastAPI - Swagger UI

127.0.0.1:8000/docs/

FastAPI 0.1.0 OAS 3.1

/openapi.json

default

- GET /students Get Students
- POST /students Create Student
- PUT /students/{student_id} Update Student
- DELETE /students/{student_id} Delete Student

Schemas

HTTPValidationError > Expand all object

Student > Expand all object